

A Minor Project Synopsis

On

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Smart Moist-Sense: Intelligent IoT-Based Soil Moisture Monitoring System

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Under the guidance of – Dr. Swati Vashisht

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

IILM UNIVERSITY, GREATER NOIDA, INDIA

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Date: 26-03-2026

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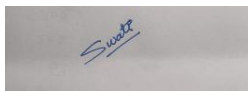
Project Team Details

Smart Moist-Sense: Intelligent IoT-Based Soil Moisture Monitoring System
Project ID: BTP2CSE023 | Session: 2025–26

Project Team	
Team Member 1	MOHD UBAID (2410030326)
Team Member 2	Shantam (2410030356)
Team Member 3	Keshav Kumar (2410030281)
Team Member 4	Divyanshu Ranjan Karn (2410030313)
Team Member 5	Nitesh Raj Singh (2410030254)

Under the Guidance of
Prof. Dr. Swati Vashisht

Signature of Guide: _____



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1. Introduction

Agriculture is the backbone of India's economy, employing nearly 42% of the workforce and contributing significantly to the national GDP. Yet water — the most critical input for crop growth — is being consumed at an unsustainable rate. Traditional flood irrigation wastes up to 60% of applied water due to the absence of real-time soil condition data. Farmers irrigate on fixed schedules or intuition, resulting in either over-irrigation (root rot, waterlogging, nutrient leaching) or under-irrigation (wilting, yield loss, plant stress).

Soil moisture — the volumetric water content in the soil relative to its total pore volume — is the single most actionable parameter for irrigation management. When monitored continuously and intelligently, it enables farmers and horticulturists to irrigate precisely when and where needed, reducing water waste while improving crop health and yield consistency.

The Internet of Things (IoT) has created an affordable and accessible pathway for such intelligent monitoring. With low-cost microcontrollers like the ESP32 and capacitive soil moisture sensors, it is now feasible to build automated, real-time sensing systems that relay data to a desktop dashboard for live visualization, alert generation, historical analysis, and predictive forecasting — at a fraction of the cost of commercial SCADA systems.

Smart Moist-Sense is exactly such a system. The ESP32 microcontroller reads analog voltage from a capacitive soil moisture sensor at one-second intervals, classifies the reading (Dry / Moderate / Wet) using configurable thresholds, activates corresponding LED indicators, and transmits structured JSON data over USB serial to a Python desktop dashboard. The dashboard provides real-time graphs, status classification, dry-alert counting, CSV logging, and a short-term moisture trend prediction — forming a complete, end-to-end precision irrigation support tool.

1.1 Background

Conventional soil monitoring is performed manually by agronomists — a slow, labour-intensive process that is inherently delayed. By the time a moisture deficit is detected manually, significant crop stress may have already occurred. The ESP32 microcontroller, with its dual-core 240 MHz processor, built-in Wi-Fi, and 12-bit ADC, combined with capacitive sensors (more accurate and durable than resistive variants), and visualized through a Python Tkinter + Matplotlib dashboard, demonstrates how modern affordable technology can replace outdated manual practices in agriculture.

1.2 Scope of the Project

This minor project implements a prototype-level IoT soil moisture monitoring system. It continuously monitors a single soil moisture parameter, classifies it into three actionable irrigation states, and delivers data to a visually rich PC dashboard via USB serial communication. The prototype is modular and extensible —

temperature, humidity, and pH sensors, as well as Wi-Fi and Telegram notification support, can be added in future iterations with minimal changes.

2. Problem Statement

Despite water's critical role in agriculture, soil moisture management across small and medium-scale farming in India remains largely reactive and imprecise. The absence of affordable, real-time monitoring tools forces farmers to depend on experience rather than objective data — leading to irrigation inefficiencies with significant consequences. According to the Food and Agriculture Organization (FAO), agriculture accounts for approximately 70% of global freshwater withdrawals. In India, over-irrigation in rice and wheat cultivation alone depletes billions of litres of groundwater annually.

Existing commercial soil moisture monitoring solutions — such as industrial SCADA systems or high-end capacitive sensor arrays — are cost-prohibitive for individual farmers, small nurseries, research labs, or academic institutions. A complete commercial monitoring node can cost thousands of rupees, making widespread deployment economically unjustifiable. Furthermore, even where sensors exist, raw data without context, classification, alerts, or historical trending has limited practical value. What is needed is not just sensing, but intelligent sensing — a system that interprets data, raises alerts, logs history, and supports informed irrigation decisions.

2.1 Key Challenges Identified

- Absence of real-time, affordable soil moisture monitoring tools for small-scale and academic use.
- Dependency on manual, infrequent soil inspection leading to delayed and often inaccurate irrigation decisions.
- Over-irrigation and under-irrigation causing crop yield loss, root diseases, soil degradation, and groundwater depletion.
- Commercial IoT monitoring solutions are cost-prohibitive for student researchers and smallholder farmers.
- Lack of intuitive visual dashboards presenting soil data in an instantly actionable format.
- No historical data logging or trend analysis in existing low-cost DIY systems.
- No automated alert mechanisms to notify users when moisture crosses safe thresholds.

3. Literature Review

A review of existing academic work in IoT-based soil moisture monitoring and smart irrigation reveals strong foundational research, but also significant gaps — particularly in accessibility, local dashboard design, offline capability, and student-level ML integration. The table below summarises key references and their limitations that this project directly addresses.

#	Reference	Contribution	Gap / Limitation
1	Gondchawar & Kawitkar (2016) IEEE	IoT-based smart agriculture using sensors and wireless networks	Proved feasibility of low-cost IoT for soil monitoring; limited to basic threshold alerts — no dashboard or ML.
2	Patil et al. (2020) Elsevier	Real-time soil moisture monitoring with ESP8266 and ThingSpeak	Used cloud logging but lacked local dashboard and offline capability; no predictive analytics.
3	Keswani et al. (2019) Springer	Adapting weather conditions with IoT-based smart irrigation	Incorporated weather data but required expensive hardware; not student-level reproducible.
4	Prathibha et al. (2017) IEEE	IoT-based monitoring system for smart agriculture	Demonstrated multi-sensor integration but used proprietary platforms; no open-source dashboard.
5	Kumar et al. (2021) MDPI Sensors	Machine learning for soil moisture prediction in precision farming	Validated ML-based forecasting but required large datasets; this project uses lightweight linear regression suited for 2nd-year level.

3.1 Research Gaps Addressed by This Project

- **No Offline Dashboard:** Most existing systems rely on cloud platforms (ThingSpeak, Blynk) requiring constant internet. Smart Moist-Sense uses a Python-based local dashboard with zero cloud dependency — fully functional offline.
- **No Local AI/ML:** Published systems do not integrate predictive analytics at the local edge level. This project embeds a simple but meaningful linear regression model directly in the dashboard for 10-minute moisture forecasting.
- **High Cost Barrier:** Commercial and research-grade systems use expensive hardware. The entire Smart Moist-Sense prototype is buildable under ₹500 using widely available components.
- **Limited Educational Value:** Existing implementations use proprietary platforms with minimal documentation. This project is fully open-source, with every component — from firmware to dashboard — designed to be learnable and viva-explainable.

4. Objectives

The primary objective is to design, develop, and demonstrate a fully functional prototype of an intelligent, IoT-based soil moisture monitoring system that is low-cost, real-time, and practically deployable for agricultural, horticultural, and academic research use.

4.1 Primary Objectives

1. To develop an ESP32-based sensor node that continuously reads analog soil moisture data from a capacitive sensor and classifies it into actionable states (Dry, Moderate, Wet) using configurable threshold logic.
2. To establish reliable USB serial communication between the ESP32 and a Python desktop dashboard using PySerial, enabling real-time JSON data streaming at 9600 baud.
3. To design and develop a Python dashboard (Tkinter + Matplotlib) showing live moisture readings, a scrolling real-time graph, soil status classification, dry-alert count, and online/offline sensor detection.
4. To implement persistent CSV-based data logging with timestamped readings for subsequent offline analysis.
5. To validate the complete system in the Wokwi online simulator before deploying to physical hardware.

4.2 Secondary Objectives

6. To incorporate a basic linear regression (NumPy) predictive module forecasting moisture levels for the next 10 minutes — overlaid on the live graph as a dashed trend line.
7. To implement sensor online/offline status detection — flagging automatically when no serial data is received within a timeout window.
8. To design the architecture with modularity so additional sensors (temperature, humidity, pH) and wireless features (Wi-Fi, MQTT, Telegram) can be added in future iterations.
9. To prepare a complete B.Tech minor project report and optionally a research paper for submission to an academic conference in IoT or smart agriculture.

5. Methodology

The development follows a structured, iterative, phase-by-phase approach progressing from simulation to hardware, and from basic sensing to intelligent analysis. The system is conceptually divided into three layers: the Sensing Layer, the Communication Layer, and the Visualization & Analysis Layer.

5.1 System Architecture

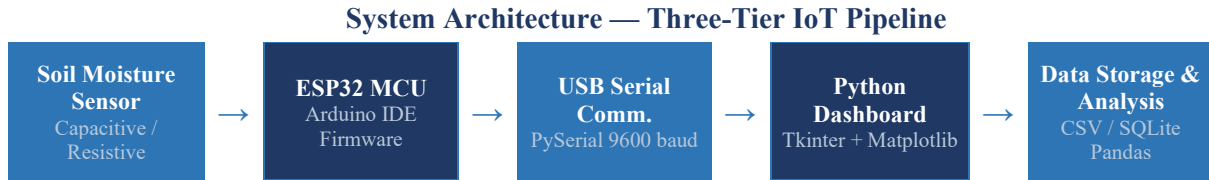


Figure 1: End-to-end data flow from physical sensor to visualization dashboard

The Sensing Layer consists of the ESP32 microcontroller and capacitive soil moisture sensor. The sensor outputs an analog voltage (0–3.3V) proportional to water content, read by the ESP32's 12-bit ADC (0–4095). The Communication Layer transmits structured JSON data packets from ESP32 to PC over USB serial at 9600 baud. The Visualization & Analysis Layer is the Python dashboard that ingests the stream, decodes JSON, updates the live graph, classifies status, logs CSV, and runs the ML prediction module.

5.2 Phase 1 — Wokwi Simulation

All hardware and firmware logic was first fully implemented in the Wokwi online ESP32 simulator, using a potentiometer to simulate analog soil moisture sensor output and three LEDs controlled by threshold logic. This phase validated all firmware logic, JSON serial output formatting, and threshold boundaries before any physical hardware was procured or assembled.

5.3 Phase 2 — ESP32 Firmware (Arduino IDE)

The firmware written in C++ performs the following cycle continuously: initialise ADC pin and serial port → read 12-bit ADC value → map to moisture percentage → apply threshold logic → activate appropriate LED → format JSON string → transmit via serial → delay 1 second. JSON format: {"moisture": 72.4, "status": "WET", "raw": 2980}. Error handling covers ADC failures and serial buffer overflows.

Threshold-Based Soil Classification Logic

Moisture Value (ADC)	Range (%)	Classification	LED / Alert
0 – 1500	< 36%	DRY Δ	RED LED ON
1501 – 2800	36% – 68%	MODERATE ~	YELLOW LED
2801 – 4095	> 68%	WET $\checkmark$	GREEN LED ON

Figure 2: ESP32 ADC-based threshold classification for irrigation decisions

5.4 Phase 3 — Python Dashboard

Built with Tkinter (GUI) and Matplotlib (charting), the dashboard provides: (1) real-time moisture percentage display with large colour-coded status text; (2) a live scrolling line graph showing the last 60 readings; (3)

dry-alert counter incrementing on each DRY classification; (4) online/offline sensor indicator with serial timeout detection; and (5) background CSV logger writing to `moisture_log.csv`. PySerial reads the COM port in a background thread using a thread-safe queue to keep the GUI responsive.

5.5 Phase 4 — ML Prediction Module

A lightweight predictive module using NumPy's `polyfit` (linear regression) is trained on the last 30 readings in the in-memory buffer. It produces a short-term 10-minute moisture trend forecast, displayed as a dashed overlay on the live graph. This gives users advance warning of approaching DRY conditions — enabling proactive rather than reactive irrigation. Linear regression was chosen for its simplicity, interpretability, and ease of explanation in a viva setting.

6. Tools & Technologies

6.1 Hardware Components

Component / Model	Role	Specification / Notes
ESP32 DevKit V1	Microcontroller	Dual-core 240 MHz, built-in Wi-Fi & BT, 12-bit ADC
Capacitive Soil Moisture Sensor v1.2	Primary Sensor	Analog output, corrosion-resistant, 3.3V–5V supply
Red LED + 220Ω Resistor	DRY Alert Indicator	Visual alert when moisture below DRY threshold
Green LED + 220Ω Resistor	WET Indicator	Visual confirmation when moisture above WET threshold
Yellow LED + 220Ω Resistor	Moderate Indicator	Visual indicator for moderate moisture range
USB Type-B Cable	Serial Communication	Data transfer ESP32 → PC at 9600 baud
Breadboard + Jumper Wires	Prototyping	Circuit assembly without soldering
5V USB Power / PC USB	Power Supply	Powers ESP32 and sensor circuitry

6.2 Software Tools and Platforms

Tool / Library	Description and Role
Arduino IDE (v2.x)	Firmware development for ESP32 — reads ADC, applies thresholds, outputs JSON via serial
Python 3.x	Core language for dashboard logic, serial communication, and data processing
Tkinter	GUI framework for desktop dashboard — real-time status display and layout
Matplotlib	Live graph plotting of moisture percentage over time with scrolling window
PySerial	Serial communication library — reads ESP32 JSON output from USB COM port
Pandas	CSV data logging, historical moisture reading storage and trend analysis
NumPy / scikit-learn	Basic linear regression for short-term moisture prediction (ML feature)
Wokwi Simulator	Online ESP32 simulation — full circuit testing before physical hardware

6.3 Communication Protocol

USB Serial (UART) at 9600 baud is the primary protocol for real-time data transfer between the ESP32 and the Python dashboard. Data is formatted as newline-delimited JSON strings — lightweight, human-readable, and easily parsed. For future wireless deployment, MQTT over Wi-Fi (HiveMQ / Mosquitto broker) will be explored, along with HTTP REST APIs for cloud integration with ThingSpeak or Firebase.

7. Expected Outcomes

Upon successful completion, Smart Moist-Sense will deliver the following concrete products and outcomes. The focus is on tangible, working deliverables rather than theoretical outputs.

Deliverable	Description
Wokwi Simulation	Complete ESP32 circuit simulation with potentiometer, LEDs, threshold logic, and serial JSON output — verifiable without physical hardware
ESP32 Hardware Prototype	Fully assembled and tested circuit: ESP32 + capacitive soil moisture sensor + 3 LEDs with correct threshold wiring
Python Desktop Dashboard	Real-time Tkinter GUI with live Matplotlib graph, DRY/MODERATE/WET status, dry-alert counter, and ONLINE/OFFLINE sensor detection
CSV Data Logger	Automated timestamped logging to moisture_log.csv — stores every reading for offline historical analysis
ML Prediction Module	Linear regression model (NumPy) trained on last 30 readings — predicts next 10-minute moisture trend overlaid on the live graph
Arduino Firmware (.ino)	Clean, commented C++ firmware reading ADC, classifying soil, driving LEDs, and transmitting JSON via Serial at 9600 baud
Project Report	Complete B.Tech minor project documentation covering all phases — submitted as per IILM University academic requirements
GitHub Repository	Open-source code repository with README, setup instructions, circuit diagram, and simulation link

7.1 Skills Gained

- Embedded systems programming (ESP32, C++, Arduino IDE, ADC interfacing)
- IoT communication protocols (Serial/UART, JSON, with foundation for MQTT and HTTP)
- Python full-stack development (Tkinter GUI, Matplotlib, PySerial, Pandas, NumPy)
- Basic ML application — linear regression for time-series prediction
- Complete project lifecycle: concept → simulation → hardware → dashboard → documentation

7.2 Social and Environmental Impact

- Reduction in water wastage through data-driven, threshold-based irrigation — contributing to SDG 6 (Clean Water) and SDG 2 (Zero Hunger).
- Sub-₹500 total hardware cost makes precision soil monitoring accessible to smallholder farmers and student researchers alike.
- Open-source codebase and detailed documentation serve as a replicable learning resource for IoT and AIML students.

8. Applications

Smart Moist-Sense has practical applications across multiple domains owing to its low cost, modular design, and real-time intelligent monitoring capability.

8.1 Precision Agriculture & Irrigation Management

Direct deployment in fields, farms, and nurseries to trigger irrigation only when soil is classified DRY — eliminating guesswork and saving water. Especially valuable for high-value crops (vegetables, medicinal herbs, strawberries) where moisture consistency directly impacts yield quality.

8.2 Greenhouse & Controlled Environment Agriculture

Multi-node deployment within greenhouses for zone-wise moisture tracking. The ML prediction module enables pre-emptive drip system activation before DRY conditions are reached — maximising yield per unit water.

8.3 Academic Research & IoT Education

An ideal end-to-end learning platform for students of CSE, ECE, and Agriculture Engineering covering embedded programming, IoT communication, Python GUI development, and basic ML — all in one cohesive project.

8.4 Urban Gardening & Smart Homes

Affordable monitoring for balcony gardens, terrace farms, and indoor plants. With the optional Telegram notification module (future scope), users receive mobile alerts when plants need watering — even when away from the PC.

8.5 Disaster Preparedness & Soil Health Research

Long-term CSV data logging supports soil health studies, drought early-warning research, and climate-resilient farming analysis — providing a free, open-source data collection node for institutional research.

9. Conclusion and Future Work

9.1 Conclusion

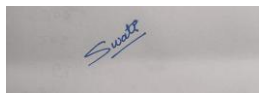
Smart Moist-Sense successfully demonstrates that intelligent, real-time soil moisture monitoring is achievable at sub-₹500 hardware cost using an ESP32, a capacitive sensor, and a Python dashboard. The system addresses a clearly defined and practically significant problem — imprecise, non-data-driven irrigation — with a technically sound, fully implementable solution. The threshold-based classification, live visualization, dry-alert tracking, CSV logging, and linear regression predictive module together form a complete precision irrigation support tool that is both academically credible and practically useful.

The iterative methodology — from Wokwi simulation to physical hardware to intelligent dashboard — ensures the project is reproducible, well-validated, and viva-ready. Every component is open-source, well-documented, and appropriate for a 2nd-year B.Tech student to implement, understand, and explain confidently. The project is strongly aligned with UN SDG 2 (Zero Hunger), SDG 6 (Clean Water and Sanitation), and SDG 9 (Industry, Innovation and Infrastructure).

9.2 Future Work

- Wi-Fi Integration: Replace USB serial with Wi-Fi (ESP32's built-in module) and MQTT protocol to enable wireless, multi-node monitoring without a physical tether to a PC.
- Telegram Bot Notifications: Push real-time DRY alerts to a Telegram chat using the python-telegram-bot library — enabling remote mobile monitoring.
- Multi-Sensor Expansion: Integrate DHT22 (temperature and humidity) and a pH sensor module to provide a full soil health dashboard — not just moisture.
- Cloud Dashboard: Migrate from local CSV to ThingSpeak or Firebase for persistent cloud storage and a publicly accessible web dashboard with historical charts.
- Improved ML Model: Upgrade from linear regression to LSTM (Long Short-Term Memory) neural networks once sufficient historical data (weeks of readings) has been logged, enabling more accurate non-linear moisture trend prediction.
- Mobile Application: Develop a lightweight Android companion app (Flutter or MIT App Inventor) displaying live moisture readings and push notifications from the ESP32 Wi-Fi node.

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Date: 26-03-2026